DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHEME & SYLLABUS

First Semester / Second Semester

(From the Academic Year 2024-25)

DEPARTMENT	**Computer Science and Engineering**					
Course Code	***24CS111/ 24CS211***	***Total Credits***	***4***	***Course Type***	***Engineering Science Course***	
Course Title	***C Programming*** *(Elective 2 - Introduction to Programming)*					
Teaching Learning Process		***Contact Hours***	***Credits***	***Assessment in Weightage and marks***		
	Lecture	**39**	**3**		**CIE**	**SEE**	**Total**
	Tutorial	**0**	**0**	***Weightage***	***40 %***	***60 %***	***100 %***
	Practical	**26**	**1**	***Maximum Marks***	***40 Marks***	***60 Marks***	***100 Marks***
	Total	**65**	**4**	***Minimum Marks***	***20 marks***	***25 marks***	***45 Marks***

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20+25 or 21+24)

COURSE PREREQUISITE: NIL.

COURSE OBJECTIVES:

Sl. No.	Course Objectives
1	Learn the basic principles and concepts of problem solving.
2	Apply the constructs of 'C' programming language for problem solving.
3	Use 'C' programming language to develop simple applications.

COURSE OUTCOMES (COs)

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Describe the concepts of problem solving with algorithmic approach and comprehend the basic constructs of 'C' language.	L2
CO2	Apply the knowledge of operators, expressions and I/O function for solving problems.	L3
CO3	Use branching and looping statements in problem solving.	L3
CO4	Solve problems using arrays and strings.	L3
CO5	Apply the concepts of pointers, modular-programming approach and structures in developing programs.	L3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Content / Syllabus:

UNIT No.	Content	Hours Lecture
1	**Introduction to Computing:** Introduction, Components of a Computer, Concepts of hardware and Software, Art of Programming through Algorithms and Flowcharts. **Overview of C:** History of C, Importance of C, Sample Programs, Basic Structure of C Programs, Programming Style, Executing a C Program. **Constants, Variables and Data Types:** Introduction, Character Sets, C	08

	Tokens, Keywords and Identifiers, Constants, Variables, data Types, Declaration of Variables, Assigning values to Variables, Defining Symbolic Constants.	
2	**Operators and Expressions:** Introduction, Arithmetic Operators, Relational Operators, Logical Operators, Assignment Operators, Increment and Decrement Operators, Conditional Operators, Bitwise Operators, Special Operators, Arithmetic Expressions, Evaluation of Expressions, Precedence of Arithmetic Operators, Some computation Problems, Type Conversion in Expressions, Operators Precedence and Associativity. **Managing Input and Output Operations:** Introduction, Reading a Character, Writing a Character, Formatted Input, Formatted output.	07
3	**Decision Making and Branching:** Introduction, Decision making with 'if' Statement, Simple 'if' Statement, The 'if…else' Statement, Nesting of 'if…else' Statements, The 'else if' Ladder, The 'switch' Statement, The ? : Operator, the 'goto' Statement. **Decision Making and Looping:** Introduction, The 'while' Statement, The 'do' Statement, The 'for' Statement, Jumps in Loops.	08
4	**Arrays:** Introduction, One-Dimensional Arrays, Declaration of One-Dimensional Arrays, Initialization of One-Dimensional Arrays, Two-Dimensional Arrays, Initialization of Two-Dimensional Arrays. **Character Arrays and Strings:** Introduction, Declaring and Initializing String Variable, Reading Strings from Terminal, Writing Strings to Screen.	08
5	**Pointers:** Introduction, Understanding Pointers, Accessing the Address of a Variable, Declaring Pointer Variables, Accessing a Variable through its Pointer. **User-Defined Functions:** Introduction, Need for User-Defined Functions, A Multi-Function Program, Elements of User-Defined Functions, Defining Functions, Return Values and Their Types, Function Calls, Function Declaration, Category of Functions. **Structures:** Introduction, Defining a Structure, Declaring Structure variables, Accessing Structure Members, Structure Initialization, Copying and comparing Structure Variables, Operations of Individual members.	08

Text Books:

Sl. No.	Author/s	Title	Publisher Details
1	E Balagurusamy	Programming with ANSI C	8th Edition, 2019, Tata McGraw Hill Publications

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	R. G Dromey	How to solve it by Computer	Pearson Education India 2015
2	Brian W. Kernighan and Dennis M. Ritchie	The C Programming Language	2nd Edition, PHI, 2012
3	Herbert Schildt	C: The Complete Reference	Paperback - 1, 4th Edition, July 2017

Web Resources:

Sl. No.	Web Link
1	https://nptel.ac.in/courses/106104128
2	https://nptel.ac.in/courses/106/105/106105171/

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	-	-	-	-	-	-	-	-	-
CO4	3	3	3	-	-	-	-	-	-	-	-	-
CO5	3	3	3	-	-	-	-	-	-	-	-	-

High – 3, Medium – 2, Low – 1

LIST OF PROGRAMS FOR PRACTICAL SESSIONS

Develop an algorithm/draw a flowchart, write a ‘C’ program and execute the following:

Weeks	List of Programs	No. of Hours
1	(i) To accept two numbers and perform basic arithmetic operations as follow: +, -, *, /, % (ii) To find the volume of a Sphere using the formula, Volume = $4 / 3\ \Pi\ r^3$ (iii) To find the area of a Triangle with 3 sides given using formula, Area = $\sqrt{(s(s-a)\ (s-b)\ (s-c))}$ where a, b and c are lengths of sides of Triangle, and s = (a+b+c) / 2	2
2	(i) Given the values of the variables x, y and z, write a program to rotate their values such that x has the value of y, y has the value of z, and z has the value of x. (ii) To read floating-pointing number and then displays the right-most digit of the integral part of the number. (iii) To read floating-pointing number, separate and displays the integral and decimal part of the given. (iv) To print the size of various data types in ‘C’ programming language using ‘sizeof’ operator.	2
3	(i) Using ‘ternary’ operator, check whether a given number is Positive or Negative. (ii) Using ‘ternary’ operator, to check whether a given Year is a Leap Year or Not. (iii) Using ‘ternary’ operator, find the largest of three numbers. (iv) Using Bitwise-AND (&) operator, check whether a given number is Odd or Even.	2
4	(i) To illustrate the use of increment operator (postfix & prefix). (ii) To illustrate the use of decrement operator (postfix & prefix). (iii) To perform the following using bitwise operators: c = a & b ; d = a | b ; e = ~a f = a >> n; g = a << n; h = a ^ b	2
5	(i) To determine whether a given number is ‘Odd’ or ‘Even’ and print the message NUMBER IS EVEN or NUMBER IS ODD with and without using ‘else’ statement. (ii) To determine whether a given number is ‘Positive’, Negative’ or	2

	'ZERO' and print the message "NUMBER IS POSITIVE", "NUMBER IS NEGATIVE" or "NUMBER IS ZERO" using nested 'if' statement. (iii) To compute all the roots of a quadratic equation by accepting the non-zero coefficients. Print appropriate messages.	
6	(i) To generate an electricity bill by accepting meter number of the consumer, number of units consumed and print out the detail charges for the below scenario: An electricity board charges the following rates for the use of electricity: ➢ for the first 200 units 80 paise per unit ➢ for the next 100 units 90 paise per unit ➢ beyond 300 units Rs 1 per unit All users are charged a minimum of Rs. 100 as meter charge. If the total amount is more than Rs. 400, then an additional surcharge of 15% of total amount is charged. (ii) A Telecommunication department charges the following tariffs for the use of Telephone-Line: ➢ For the first 100 units 95 paise per unit ➢ For the next 75 units Rs. 1.50 per unit ➢ Beyond 175 units Rs 2.85 per unit. All users are charged a minimum of Rs. 75 as service charge. If the total amount is more than Rs 500, then an additional surcharge of 12.5% of total amount is charged. Write a program to read the name of the user, number of units consumed and print out the charges.	2
7	(i) To input month number and display its respective month in word. (ii) To simulates a simple calculator to perform the basic arithmetic operations (Consider the operators +, -, x, / and % using 'switch' statement) (iii) To check whether a given alphabet is vowel or consonant using 'switch' statement.	2
8	(i) To sum Natural, Odd and Even numbers up to 'N'. (ii) To generate and print the first 'N' Fibonacci numbers such that Fn = F(n-1) + F(n-2) where n>2. A Fibonacci sequence is defined as "the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence". (iii) To find the sum of individual digits of a positive integer number reducing into single digit. (iv) To reverse a given four-digit integer number and check whether it is a palindrome or not. Output the given number with suitable message.	2
9	(i) To determine whether a given number is Prime or not. (ii) To generate and print all the prime numbers between range N1 and N2, where 'N1' and 'N2' are value supplied by the user. (iii) To find the value of cos(x) using the series, 1 - x2/2! + x4/4! - x6/6! + x8/8! - ... up to N terms accuracy (With and without using in-built function) (iv) To find the value of sin(x) using the series, x - x3/3! + x5/5! - x7/7! + x9/9! - ... up to N terms accuracy (With and without using in-built function)	2
10	(i) To input N Real Numbers and to find mean, variance and standard deviation using appropriate formula. (ii) To input N integer numbers into a single dimension array, sort them in to ascending order using "BUBBLE SORT" technique, and then to print both the given array and the sorted array with suitable headings.	2

	(iii) To input N integer numbers in into a single dimension array, and then to perform “LINEAR SEARCH” for a given Key integer number and report success or failure in the form of a suitable message.	
11	(i) To input N integer numbers in ascending order into a single dimension array, and then to perform “BINARY SEARCH” for a given Key integer number and report success or failure in the form of a suitable message. (ii) To perform addition and subtraction of two matrices after checking their compatibility and print both input & output matrices with suitable headings. Use user- defined functions to read and print the matrices.	2
12	(i) To read a matrix A (M x N), find the TRANSPOSE of the given matrix and output both the input matrix and the transposed matrix. Use user-defined functions to find the TRANSPOSE. (ii) To find the TRACE and NORM of a given matrix A (M x N) by checking the compatibility and print both input & output matrices with suitable headings. Use user-defined functions to find their TRACE and NORM.	2
13	**Laboratory Test**	2

<table>
<tr><td>DEPARTMENT</td><td colspan="7">Computer Science and Engineering</td></tr>
<tr><td>Course Code</td><td>24CS112/ 24CS212</td><td>Total Credits</td><td>4</td><td>Course Type</td><td colspan="3">Engineering Science Course</td></tr>
<tr><td>Course Title</td><td colspan="7">Python Programming
(Elective 2 - Introduction to Programming)</td></tr>
<tr><td rowspan="5">Teaching Learning Process</td><td></td><td>Contact Hours</td><td>Credits</td><td colspan="4">Assessment in Weightage and marks</td></tr>
<tr><td>Lecture</td><td>39</td><td>3</td><td></td><td>CIE</td><td>SEE</td><td>Total</td></tr>
<tr><td>Tutorial</td><td>0</td><td>0</td><td>Weightage</td><td>40 %</td><td>60 %</td><td>100 %</td></tr>
<tr><td>Practical</td><td>26</td><td>1</td><td>Maximum Marks</td><td>40 Marks</td><td>60 Marks</td><td>100 Marks</td></tr>
<tr><td>Total</td><td>65</td><td>4</td><td>Minimum Marks</td><td>20 marks</td><td>25 marks</td><td>45 Marks</td></tr>
</table>

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

COURSE PREREQUISITE: Knowledge of programming.

COURSE OBJECTIVES:

Sl. No.	Course Objectives
1	Introduce the syntax and semantics of Python Programming Language.
2	Write python functions to facilitate code reusability.
3	Illustrate the concepts of Strings, lists, tuples and dictionaries.

COURSE OUTCOMES (Cos)

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Understand the basics of digital computer and various problem solving strategies	L2
CO2	Illustrate the basic constructs of Python Programming Language.	L3
CO3	Implement the concepts of built-in and user-defined functions in Python.	L3
CO4	Implement the concepts of Strings and Lists in Python.	L3
CO5	Implement methods to create tuples and dictionaries. Understand the basic concepts of object oriented programming.	L3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Content / Syllabus:

UNIT No.	Content	Hours Lecture
1	**Introduction to Digital Compute**r: Introduction, Von Neumann Concept, Storage, Programming languages, Translators, Hardware and Software, Operating Systems. **Problem Solving Strategies**: Problem Analysis, Algorithms, Flow Charts, Examples of Algorithms and Flow Charts.	07
2	**Introduction to Python**: Introduction, Python Overview, Getting Started with Python, Comments, Python Identifiers, Reserved Keywords, Variables, Standard Data Types, Operators, Statement and Expression, String Operations, Boolean Expressions, Control Statements, Iteration – while Statement, Input from Keyboard.	08
3	**Functions:** Introduction, Built-in Functions: Type conversion, Type Coercion, Mathematical Function, Date and Time, dir() Function, help() Function, Composition of Functions, User Defined Functions, Parameters and Arguments, Function Calls, The return statement.	08
4	**Strings and Lists:** Strings**:** Compound datatype, len function, string slices, Immutable strings, string traversal, escape characters, string formatting operator, string formatting functions. List**:** Values and accessing elements, Mutable lists, Traversing a List, Deleting elements from List, Built-in list Operators, Built-in list Methods.	08
5	**Tuples and Dictionaries:** Tuples: Creating Tuples, accessing values in Tuples, Immutable Tuples, Tuple Assignment, Basic Tuple operations, Built-in Tuple functions. Dictionaries: creating a Dictionary, accessing values, Updating Dictionary, Deleting Elements. **Classes and Objects:** Overview of OOP, Class Definition, Creating Objects.	08

Text Books:

Sl. No.	Author/s	Title	Publisher Details
1	E Balagurusamy	Introduction to Computing and Problem-Solving Using Python	McGraw Hill Education India Private Limited; First Edition (1 July 2017), ISBN-10: 9352602587 ISBN-13: 978-9352602582

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	R. Nageswara Rao	Core Python Programming	3rd Edition, Dream tech Press, 2021
2	Wesley J. Chun,	Core Python Applications Programming	3rd Edition, Pearson Education, 2016
3	T.R. Padmanabhan	Programming with Python	1st Edition, Springer, 2016

Web Resources:

Sl. No.	Web Link
1	https://onlinecourses.swayam2.ac.in/cec24_cs11/preview
2	https://onlinecourses.nptel.ac.in/noc24_cs113/preview

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	-	-	-	-	-	-	-	-	-
CO4	3	3	3	-	-	-	-	-	-	-	-	-
CO5	3	3	3	-	-	-	-	-	-	-	-	-

High – 3, Medium – 2, Low – 1

LIST OF PROGRAMS FOR PRACTICAL SESSIONS

Develop a Python program and execute the following:

Weeks	List of Programs	No. of Hours
1.	a) Program to find the square root of a number. b) Program to find the area of a rectangle, triangle and circle.	2
2.	a) Program to swap the values of two variables. b) Program to convert Celsius to Fahrenheit and vice versa.	2
3.	a) Program to find whether the number is even or odd. b) Program to find the largest among three numbers.	2
4.	a) Program to check whether the given year is Leap year or not. b) Program to check whether the number is positive, negative or Zero.	2
5.	a) Program to display Fibonacci number till ‘n’ terms. b) Program to check whether the given number is prime or not.	2
6.	a) Program to find the factorial of a number using user defined function. b) Program to find the sum of digits of a given number using user defined function.	2
7.	a) Program to find the GCD and LCM of two integers using user defined function. b) Program to find ASCII value of a character using user defined function.	2
8.	a) Program to check whether the given string is palindrome or not. b) Program to find i) average of elements in the list	2

	ii) Sort the elements in the list iii) Remove given element using built in list methods.	
9.	a) Program to input a tuple and to create another tuple that contains only even numbers from the given tuple. b) Program to input a integer n and to generate a dictionary that contains the pairs (i:i*i) where i range between 1 to n. print the dictionary in suitable format.	2
10.	Program to define a class called Rectangle that takes the length and breadth as its parameter and a method to compute its perimeter.	2
11.	a) Program to demonstrate classes and their attributes. b) Program to demonstrate various tuple functions and operations.	2
12.	Program to demonstrate various dictionary functions and operations.	2
13.	**Laboratory Test**	2

<table>
<tr><td>DEPARTMENT</td><td colspan="7">Computer Science and Engineering</td></tr>
<tr><td>Course Code</td><td>24CS113/ 24CS213</td><td>Total Credits</td><td>4</td><td>Course Type</td><td colspan="3">Engineering Science Course</td></tr>
<tr><td>Course Title</td><td colspan="7">Object Oriented Programming using JAVA
(Elective 2 - Introduction to Programming)</td></tr>
<tr><td rowspan="5">Teaching Learning Process</td><td></td><td>Contact Hours</td><td>Credits</td><td colspan="4">Assessment in Weightage and marks</td></tr>
<tr><td>Lecture</td><td>39</td><td>3</td><td></td><td>CIE</td><td>SEE</td><td>Total</td></tr>
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</table>

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20+25 or 21+24)

COURSE PREREQUISITE: Knowledge of programming.

COURSE OBJECTIVES:

Sl. No.	Course Objectives
1	Understand the basic concepts of Object-Oriented Programming and apply them in problem solving.
2	Gain a clear understanding of Java programming and set up Java JDK environment to create, debug and run simple Java programs.
3	Explore the concepts of reusability and exception-handling mechanism in Java.
4	Study how multi-threaded applications can be designed and developed in Java.

COURSE OUTCOMES (COs)

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Explain the concepts of Object Oriented Programming and JAVA.	L2
CO2	Design the JAVA applications using classes and methods	L3
CO3	Incorporate the code reusable properties with Inheritance.	L3
CO4	Explore the code reusable properties with Packages and Interfaces.	L3
CO5	Implement the exception handling and multi-threaded programming mechanisms.	L3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Content / Syllabus:

UNIT No.	Content	Hours Lecture
1	**INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING AND JAVA:** Introduction, Procedure–Oriented Programming System, Object- Oriented Paradigm, Basic concepts of Object-Oriented Programming, Objects and Classes, Data Abstraction and Encapsulation, Inheritance, Dynamic Binding, Polymorphism, Message Communication, Benefits of OOP, Applications of OOP, How Java differs from C and	08

	C++. **The History and Evolution of Java:** Java's Magic: The byte code, The Java Buzzwords: Simple, Secure, Portable, Object-oriented, Robust, Multithreaded, Architecture-neutral, Interpreted, High performance, Distributed, Dynamic; Java Development Kit (JDK). **An Overview of Java**: Object-Oriented Programming: Abstraction, The three OOP Principles: Encapsulation, Polymorphism, Inheritance; Simple Java Programs; Data Types, Variables and Arrays; Operators; Control Statements.	
2	**JAVA CLASSES AND METHODS:** **Introducing Classes**: Class Fundamentals, declaring objects, Assigning Object Reference Variables, Introducing Methods, Constructors, The 'this' keyword. **A Closer Look at Methods and Classes**: Overloading Methods, Using Object as Parameters and Return Value, Access Control, Static Members, 'final'.	08
3	**CLASSES AND REUSABLE PROPERTIES:** **Inheritance**: Inheritance Basics, using super, creating a Multilevel hierarchy, When Constructors are Executed, Method Overriding.	08
4	**REUSABLE PROPERTIES (continued):** **Packages and Interfaces:** Packages, Member Access, Importing Packages, Interfaces, Default Interface Methods, Using 'static' Methods in an Interface.	07
5	**EXCEPTION HANDLING & MULTITHREADED PROGRAMMING:** **Exception Handling**: Exception-Handling Fundamentals, Exception Types, Uncaught Exceptions, using 'try' and 'catch'. **Multithreaded Programming**: The Java Thread Model, The Main Thread, creating a Thread, Obtaining a Thread's State.	08

Text Books:

Sl. No.	Author/s	Title	Publisher Details
1	E Balagurusamy	Programming with Java - A Primer	6th Edition, Tata McGraw Hill Education, 2019

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	Herbert Schildt	Java: The Complete Reference	11th Edition, Tata McGraw Hill Education, Released December 2018
2	Harvey M. Deitel, Paul J.Deitel	The Complete Java Training Course	2nd Edition, Prentice Hall, December 2001
3	Sachin Malhotra, Saurabh Choudhary	Programming in Java	2nd Edition, Oxford University Press India, 2018

Web Resources:

Sl. No.	Web Link
1	https://nptel.ac.in/courses/106/105/106105191/
2	https://cse.iitkgp.ac.in/~dsamanta/java/index.htm

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	-	-	-	-	-	-	-	-	-
CO4	3	3	3	-	-	-	-	-	-	-	-	-
CO5	3	3	3	-	-	-	-	-	-	-	-	-

High – 3, Medium – 2, Low – 1

LIST OF PROGRAMS FOR PRACTICAL SESSIONS

Weeks	List of Programs	No. of Hours
1.	a) To accept two numbers and perform basic arithmetic operations as follow: +, -, *, /, % b) Program to convert Celsius to Fahrenheit and vice versa.	2
2.	a) Program to swap the values of two variables. b) To find the volume of a Sphere using the formula, Volume = 4 / 3 Π r^3	2
3.	a) To find the area of a Triangle with 3 sides given using formula, Area = √(s(s-a) (s-b) (s-c))	2

	where a, b and c are lengths of sides of Triangle, and s = (a+b+c) / 2 b) Given the values of the variables x, y and z, write a program to rotate their values such that x has the value of y, y has the value of z, and z has the value of x.	
4.	a) Program to find whether the number is even or odd. (Use 'Ternary Operator') b) Program to find the largest among three numbers. (Use 'Ternary Operator')	2
5.	a) To determine whether a given number is 'Odd' or 'Even' and print the message NUMBER IS EVEN or NUMBER IS ODD with and without using 'else' statement. b) To determine whether a given number is 'Positive', Negative' or 'ZERO' and print the message "NUMBER IS POSITIVE", "NUMBER IS NEGATIVE" or "NUMBER IS ZERO" using nested 'if' statement.	2
6.	a) Generate prime numbers for the given range. b) Find all the roots of a quadratic equation.	2
7.	a) Print 'N' Fibonacci numbers. b) Print pyramid shape using *symbols.	2
8.	Define a class to represent a bank ACCOUNT and include the following members: I. Data Members (***States***): a) Name of Depositor b) Account number c) Type of Account d) Balance amount in the account II. Member Methods (***Behaviors***): a) To assign initial values b) To deposit an amount c) To withdraw an amount after checking for the balance d) To display name &balance III. Define ***EXECUTEACCOUNT*** class that defines main method to test above class.	2
9.	The daily maximum temperatures recorded for 5 cities during the first 6 days of January month have to be tabulated. Develop an application to read the data and find the city and day corresponding to highest temperature and lowest temperature.	2
10.	An election is contested by 5 candidates. The candidates are numbered 1 to 5 and the voting is done by marking the candidate number on the ballot paper. Develop an application to read the ballots and count the votes cast for each candidate using an array variable count. In case, a number read is outside the range 1 to 5, the ballot should be considered as a 'spoilt ballot' and the program should also count the number of spoilt ballots.	2
11.	Create a class of objects CUBE. Develop an application to read the side for	2

	three cubes and print the Volume and outer area.	
12.	Write a program to read a list containing Book Title, Book Code, Cost and Quantity interactively for min. of 10 books and produce five columns output as shown below. NAME CODE UNIT PRICE QUANTITY TOTAL PRICE Define the suitable functions and perform the transactions.	2
13.	**Laboratory Test**	2

DEPARTMENT	**Computer Science and Engineering**						
Course Code	***24HU114/ 24HU214***	***Total Credits***	***2***	***Course Type***	***Sustainability / Skill***		
Course Title	***Web Technology***						
Teaching Learning Process		***Contact Hours***	***Credits***	***Assessment in Weightage and marks***			
	Lecture	**26**	**2**		**CIE**	**SEE**	**Total**
	Tutorial	**0**	**0**	***Weightage***	***40 %***	***60 %***	***100 %***
	Practical	**0**	**0**	***Maximum Marks***	***20 Marks***	***30 Marks***	***50 Marks***
	Total	**26**	**2**	***Minimum Marks***	***10 Marks***	***13 marks***	***23 Marks***

Note: *For passing the student has to score a minimum of 23 Marks (CIE+SEE: 10+13 or 11+12)

COURSE PREREQUISITE: NIL

COURSE OBJECTIVES:

Sl. No.	Course Objectives
1	Understand the concepts and principles of Web.
2	Gain a clear understanding of HTML/XHTML.
3	Learn the concepts of Cascading Style Sheets (CSS).
4	Comprehend the essence of Java Scripting Language.
5	Explore the concepts of JavaScript and HTML Documents.

COURSE OUTCOMES (COs)

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Describe the concepts principles of Web.	L2
CO2	Design and develop web pages using HTML/XHTML and its elements.	L3
CO3	Implement the visual presentation of the web page using CSS.	L3
CO4	Build the web pages using Java Scripting Language.	L3
CO5	Design Web Applications using the features of JavaScript and HTML Documents.	L3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Content / Syllabus:

UNIT No.	Content	Hours Lecture
1	**Fundamentals of Web**: Internet, WWW, Web Browsers and Web Servers, URLs, MIME, HTTP, Security, The Web Programmers Toolbox.	05
2	**Introduction to HTML/XHTML:** Origins and Evolution of HTML and XHTML, Basic syntax, Standard structure, Basic text markup, Images, Hypertext Links, Lists, Tables, Forms, Frames.	06
3	**Cascading Style Sheets (CSS):** Introduction, Levels of style sheets, Style specification formats, Selector forms, Property value forms, Font properties, List properties, Colour, Alignment of text, The box	05

	model, Background images.	
4	**The Basics of JavaScript:** Overview of JavaScript, Object Orientation and JavaScript, Syntactic Characteristics, Primitives, Operations and Expressions, Screen Output and Keyboard Input, Control Statements.	05
5	**JavaScript and HTML Documents:** The JavaScript execution environment, The Document Object Model, Element access in JavaScript, Events and event handling.	05

Text Book:

Sl. No.	Author/s	Title	Publisher Details
1	Robert W Sebesta	Programming the World Wide Web	8th Edition, Pearson Education, 2015.

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	M. Deitel, P.J. Deitel, A. B. Goldberg	Internet & World Wide Web How to Program	3rd Edition, Pearson education, 2004.
2	Chris Bates	Web Programming Building Internet Applications	3rd Edition, Wiley India, 2006.
3	Xue Bai et al	The web Warrior Guide to Web Programming	Thomson, 2003

Web Resources:

Sl. No.	Web Link
1	https://nptel.ac.in/courses/106105084
2	https://onlinecourses.swayam2.ac.in/aic20_sp11/preview

Course Articulation:

COURSE OUTCOMES	PROGRAM OUTCOMES											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	P012
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	-	-	-	-	-	-	-	-	-
CO4	3	3	3	-	-	-	-	-	-	-	-	-
CO5	3	3	3	-	-	-	-	-	-	-	-	-

High – 3, Medium – 2, Low – 1